

Level: 1st Year

Midterm Exam

Material: Network Foundation 2

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Full name: Group : ...

Part 1: QCM (5 pts)

1. Which Wireshark feature allows us to capture only specific types of traffic before packets are captured?

- A. Display Filter
- B. Capture Filter
- C. Packet List
- D. Capture Options

2. What are the purposes of the feature capture options in Wireshark?

- A. To avoid unnecessary traffic
- B. To customize how packets are stored and handled
- C. To show only packets matching certain criteria
- D. To encrypt packets

3. What are the purposes of the "Simulation Mode" in Packet Tracer?

- A. To speed up the network
- B. To observe packet flow step-by-step
- C. To configure devices automatically
- D. To display protocol details (ARP, ICMP, DNS, etc.)

4. What does a PC need to access a file shared on a network?

- A. A driver software
- B. Proper permissions
- C. A web browser
- D. A USB cable

5. What types of cable is typically used to connect to a Router from a PC?

- A. Crossover cable
- B. Straight-through cable
- C. Fiber cable

6. What is the role of the promiscuous mode in Wireshark?

- A. Capturing only encrypted traffic
- B. Capturing only local machine traffic
- C. Blocking suspicious packets
- D. Capturing all packets on the network interface

7. What happens when two devices are in the same LAN without routing?

- A. They communicate normally
- B. They cannot communicate
- C. They share the same broadcast domain
- D. They automatically connect

8. What type of network is commonly used for printer sharing in offices?

- A. WAN
- B. LAN
- C. MAN
- D. Wi-Fi

9. Which of the following is the advantage of printer sharing?

- A. Reduced hardware costs
- B. Easier collaboration
- C. Reduced network load
- D. Reduced printing time

10. Which of the following compromises the security of file sharing?

- A. Disabling passwords
- B. Using strong authentication
- C. Sharing with everyone
- D. Saturating the network

Part 2: Exercise 01 (5.5 pts)

1- Provide the steps used to capture a packet using Wireshark ?

2- Describe the following capture and give the format of its file store ?

Time	Source	Destination	Protocol	Length	Info
1 0.000000	0.0.0.0	255.255.255.255	DHCP	114	DHCP Discover - Transaction ID 0x3d1d
2 0.070345	192.168.0.1	192.168.0.10	DHCP	112	DHCP ACK - Transaction ID 0x3d1d
3 9898282.178593	192.168.170.8	192.168.170.20	DNS	70	Standard query 0x1032 TXT google.com
4 441374633.761192	HonHaiPr_35:79:b6	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.182
5 441374633.763273	ec:cb:30:4c:dd:7d	HonHaiPr_35:79:b6	ARP	60	192.168.1.1 is at ec:cb:30:4c:dd:7d

3- What if you only need to display ICMP packets?

4- Provide the Wireshark capture if we ping from PC1 (192.168.1.1) to PC2 (192.168.1.1) ?

5- Provide the Wireshark capture. If the ping from PC1 (192.168.1.1) to 192.168.1.255 fails ?

Exercise2: (3.5 pts) Answer the following questions:

1. What are the **advantages and disadvantages** of:

○ **Static and dynamic routing?**

○ **Link-state and distance-vector routing algorithms?**

2. **Define the concept of administrative distance ?**

3. **What command is used to display the routing table?**

Exercise3: (6 pts)

Following the two routing tables, draw the network topology

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
R 10.0.0.0/8 [120/3] via 201.100.11.2, 00:00:09, GigabitEthernet0/0/1
R 192.5.5.0/24 [120/1] via 201.100.11.2, 00:00:09, GigabitEthernet0/0/1
R 192.168.1.0/24 [120/1] via 219.17.100.2, 00:00:23, GigabitEthernet0/0/2
R 199.6.13.0/24 is variably subnetted, 2 subnets, 2 masks
C   199.6.13.0/24 is directly connected, GigabitEthernet0/0/0
L   199.6.13.1/32 is directly connected, GigabitEthernet0/0/0
R 199.200.10.0/24 [120/2] via 199.6.13.2, 00:00:01, GigabitEthernet0/0/0
R 201.100.11.0/24 is variably subnetted, 2 subnets, 2 masks
C   201.100.11.0/24 is directly connected, GigabitEthernet0/0/1
L   201.100.11.1/32 is directly connected, GigabitEthernet0/0/1
R 204.204.7.0/24 [120/1] via 199.6.13.2, 00:00:01, GigabitEthernet0/0/0
R 219.17.100.0/24 is variably subnetted, 2 subnets, 2 masks
C   219.17.100.0/24 is directly connected, GigabitEthernet0/0/2
L   219.17.100.1/32 is directly connected, GigabitEthernet0/0/2
R 222.14.1.0/24 [120/2] via 201.100.11.2, 00:00:09, GigabitEthernet0/0/1
```

Router#

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, GigabitEthernet0/0/1
L 10.10.10.1/32 is directly connected, GigabitEthernet0/0/1
R 192.5.5.0/24 [120/1] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 192.168.1.0/24 [120/4] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 199.6.13.0/24 [120/3] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 199.200.10.0/24 [120/5] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 201.100.11.0/24 [120/2] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 204.204.7.0/24 [120/4] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 219.17.100.0/24 [120/3] via 222.14.1.2, 00:00:03, GigabitEthernet0/0/0
R 222.14.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 222.14.1.0/24 is directly connected, GigabitEthernet0/0/0
L 222.14.1.1/32 is directly connected, GigabitEthernet0/0/0
```

Router#